Surendhar Saravanan

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Research Profile

Ph.D. candidate in Control and Automation at IIT Delhi, working on adaptive control, distributed estimation, and multi-agent robotic systems, with theoretical analysis and experimental validation using networks of quadrotor UAVs.

Employment History

2022 – 2025 **Research Assistant (Ph.D.).** Department of Electrical Engineering, Indian Institute of Technology Delhi (IIT Delhi), New Delhi, India.

2017 – 2020 **Assistant Professor.** Thiagarajar College of Engineering (TCE), Anna University, Chennai, India.

2014 – 2015 **Assistant System Engineer.** Tata Consultancy Services Limited (TCS).

Education

2020 – Present **Ph.D., Indian Institute of Technology Delhi**.
Thesis title: *Distributed Adaptive Coverage Control for Mobile Sensor Networks: Application to a Network of Drones*. **CGPA: 9.16**

2015 – 2017 **M.E. in Instrumentation Engineering, Madras Institute of Technology, Anna University**. **CGPA: 7.85**
Thesis title: *PI-PD Controller Using Extended MPC for a Binary Distillation Column*.

2010 – 2014 **B.E. in Electronics and Instrumentation Engineering, Anna University**. **CGPA: 7.89**

Research interests

• Adaptive Control • Learning-Based Control • Distributed Estimation • Multi-Agent Systems • Optimization • Autonomous Systems • Robotics • Online Decision-Making

Experiences

- **Academic Experience:**

2020–2022 Served as a **Teaching Assistant** for postgraduate-level courses in Adaptive Control and Probability, the undergraduate-level course Basic Electrical Engineering, and the NPTEL course on Adaptive Control.

2017–2020 Gained **teaching experience** in Industrial Automation, Signal Processing and Instrumentation, and Project Management.

- **Industrial Experience:** Worked on performance and load testing using HP LoadRunner and SQL, and received training in C#.

Skills

Languages	English (fluent), Tamil (native)
Programming	Python, C++, MATLAB, ROS/ROS2, SQL, C#, LaTeX
Simulation & Tools	Gazebo, PX4 SITL, Docker, Git, Linux (Ubuntu), Visual Studio Code
Experimental Platforms	Crazyflie UAVs, OptiTrack motion capture system, ROS-based multi-robot systems

Funded Projects

2022–2025 **Research Assistant:** Title: Distributed Adaptive Coverage Control of a Network of Drones – Theory to Practice, Technology Innovation Hub (TIH) for Cobotics, IIT Delhi.
Summary: This project focused on the development of a **distributed coverage control algorithm** that drives multiple unmanned aerial vehicles (UAVs) toward optimal sensing locations while simultaneously learning the spatial distribution of variables such as the Normalized Difference Vegetation Index (NDVI) for precision agriculture and soil moisture content using multispectral and thermal imagery. The project was carried out in collaboration with the startup partner Vecros Pvt. Ltd.

2015–2016 Title: Development of Advanced Control Schemes for a Binary Distillation Column, Student Innovation Project, Centre for Technology Development and Transfer (CTDT), Anna University.
Summary: This funded project focused on the development of adaptive tuning schemes for conventional controllers, such as PID and fractional-order PID controllers, for an ethanol–water binary distillation column.

Research Publications

Publications

1. S. Surendhar, S. B. Roy, and S. Bhasin, "Online kernel adaptive coverage control with relaxed excitation condition," 2026, (Under submission).

2. S. Surendhar, S. B. Roy, and S. Bhasin, "Online kernel estimation with relaxed excitation condition," 2026, (In preparation).

3. S. Surendhar, S. B. Roy, and S. Bhasin, "Distributed adaptive coverage control with obstacle avoidance for a drone network based on collective initial excitation," *Robotics and Autonomous Systems*, vol. 194, p. 105 135, 2025, ISSN: 0921-8890. DOI: https://doi.org/10.1016/j.robot.2025.105135.

4. S. Surendhar, S. B. Roy, and S. Bhasin, "Collective initial excitation-based distributed composite adaptive coverage control with application to a network of drones," *IEEE Control Systems Letters*, vol. 8, pp. 309–314, 2024. DOI: 10.1109/LCSYS.2023.3335957.

Conferences

1. S. Surendhar, S. B. Roy, and S. Bhasin, "Obstacle avoidance adaptive coverage control using collective initial excitation," 4, 10th IFAC Conference on Networked Systems NECSYS 2025, vol. 59, 2025, pp. 61–66. DOI: https://doi.org/10.1016/j.ifacol.2025.07.045.

2. S. Surendhar, S. B. Roy, and S. Bhasin, "Collective initial excitation-based distributed adaptive coverage control for multiple drone systems: Theory to experiments," in *Poster presented at the 2023 Ninth Indian Control Conference (ICC)*, Visakhapatnam, India, 2023.

3. B. Ashok Kumar, S. Gayathri, S. Surendhar, S. Senthilrani, and R. Jeyabharathi, "Robust h-infinity controller for two degree of freedom helicopter," in *2019 IEEE International Conference on System, Computation, Automation and Networking (ICSCAN)*, 2019, pp. 1–5. DOI: 10.1109/ICSCAN.2019.8878823.

4. B. Ashok Kumar, R. Jeyabharathi, S. Surendhar, S. Senthilrani, and S. Gayathri, "Control of four tank system using model predictive controller," in *2019 IEEE International Conference on System, Computation, Automation and Networking (ICSCAN)*, 2019, pp. 1–5. DOI: 10.1109/ICSCAN.2019.8878700.

5. S. Surendhar, "Fractional order controller for the two product distillation column," in *Paper presented at NITFEAST-17 International Conference*, National Institute of Technology, Tiruchirappalli, Tiruchirappalli, India, 2017.

6. S. Surendhar, A. Ferheen, and C. Rithicka, "Advance control schemes for the binary distillation column," in *Poster presented at the 9th International Conference on Trends in Industrial Measurement and Automation (TIMA 2017)*, Chennai, India, 2017.